Robotrix 2025-26 Final Hackathon Problem Statement

Operation Sky Link

Objective

The CoppeliaSim simulation scene contains a large walled maze (15m x 15m). Inside the maze, there is a terrestrial rover ("The Mole") and a quadrotor drone ("The Hawk").

The **Mole (Rover)** is blinded; it has no onboard sensors. The **Hawk (Drone)** meanwhile has a camera attached to its base, allowing it to see the floor, and the mole on the floor. However, the hawk can only fly at a low altitude of 1.5m, meaning its camera can only see a small section of the maze at any given time.

Your objective is to guide the blind Mole from the start zone to the end zone. The Hawk must autonomously track and follow the Mole to keep it inside its camera frame. The drone must also process the video feed to track walls and a path, and send velocity commands to the Mole to navigate the maze.

Instructions

- Download the Coppeliasim world file from here.
- Download the Python script template from here.
- Do not edit the world file.
- Edit the Python script template to perform the task according to the objective.
- The codes should be neat, well structured, and well documented.

Specifications

The hazardous arena has the following measurements and properties:

- **Dimensions of Total Arena:** 15000 mm x 15000 mm.
- **Mole:** A differential drive robot. It has a **yellow** marker on top for visual tracking. It has no active sensors.
- **Hawk:** A quadcopter flying at a fixed altitude of 2.5m. It has a camera attached to its base, the camera has a resolution of 512 x 512 px, with a FOV of 90 degrees.

Constraints:

- **Communication Blackout:** You are not allowed to use global variables to share state between the Drone and the Rover. The only data sent from the Hawk to the Rover is velocity commands. The functions to send the corresponding data are already defined in the python script given. Use them to send velocity commands.
- **Anti-God Mode:** You are not allowed to use the `sim.getObjectPosition()` function to find the positions of the walls, rover's coordinates, or the end zone. You must use the sensors provided.

Score Distribution

The simulation runs for a maximum of **5 minutes**.

- **Maze completion:** 40 points for reaching the End zone
- **Tracking stability:** 20 points for tracking accuracy.
- **Wall collision penalty:** -2 points per collision with a wall (If the mole is stuck on the same wall after six seconds, that will count as a second collision, and so on for every six seconds thereafter)
- **Time Bonus:** 40 points, ticks down linearly from the 3 minute mark.

The simulation stops automatically when the Rover touches the End Zone or the time runs out. Don't worry if the timer is moving faster than real-time. It is supposed to measure simulation time to ensure fairness, and your device can simulate faster than real-time.

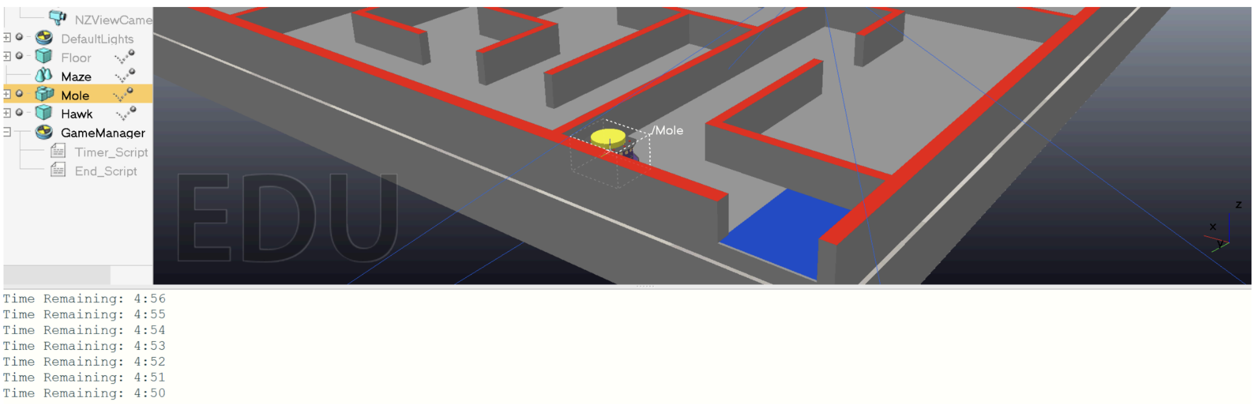
Submission Guidelines

- The submission will be via **Google Drive**. The link to the drive folder should be submitted on Unstop. Ensure that you provide view access to the folder.

- The following contents should be included in the drive folder:
  - The final Python script used for the solution. The script should be neat and readable, with comments provided wherever required.
  - A screen recording of the running simulation.
  - A text file explaining the approach used for the solution.

- Guidelines for the screen recording:

    - Software like OBS Studio (cross-platform) or Kazam (Linux only) can be used for screen recording.
    - The video must show the **Drone's Camera View** (using cv2.imshow) alongside the simulation window.
        - You must draw a bounding box around the detected Rover in the video feed to prove your vision logic is working.
    - The video should be recorded from the start of the simulation to the end.
    - The initial part of the video should briefly show the code being used for the simulation.
    - The message window of Coppeliasim should be **clearly** visible in the recorded video.



- The link to the Google Drive folder should be submitted before the deadline - **21st December 9:00 AM**. Any changes made to the folder after the deadline will lead to disqualification.

GOOD LUCK!